

Signal Processing

Exercise 3 – Fourier transform - Solution

Summer Term 2026

Task 1

a)

$$\begin{aligned} \circ x(t) &= \text{rect}\left(\frac{t}{2T_0}\right) \\ \bullet X(j\omega) &= \int_{-\infty}^{\infty} x(t) \cdot e^{-j\omega t} dt \\ &= \int_{-T_0}^{T_0} \text{rect}\left(\frac{t}{2T_0}\right) \cdot e^{-j\omega t} dt \\ &= \int_{-T_0}^{T_0} 1 \cdot e^{-j\omega t} dt \\ &= \frac{1}{-j\omega} [e^{-j\omega t}]_{-T_0}^{T_0} \\ &= \frac{1}{-j\omega} (e^{-j\omega T_0} - e^{+j\omega T_0}) \\ &= \frac{-2j \cdot \sin(\omega T_0)}{-j\omega} \\ &= \frac{2 \cdot \sin(\omega T_0)}{\omega} \cdot \frac{T_0}{T_0} \\ &= 2T_0 \cdot \text{si}(\omega T_0) \end{aligned}$$

b) Time domain:

Increment of $T_0 \rightarrow$ Increment of the width of the rectangle

Reduction of $T_0 \rightarrow$ Reduction of the width of the rectangle

Frequency domain:

Increment of $T_0 \rightarrow$ Compression of the si function

Reduction of $T_0 \rightarrow$ Elongation of the si function

Task 2

$$\begin{aligned}
 \circ x(t) &= \delta(t) = \lim_{T_0 \rightarrow 0} \left(\frac{1}{2T_0} \text{rect}\left(\frac{t}{2T_0}\right) \right) \\
 \bullet X(j\omega) &= \int_{-\infty}^{\infty} x(t) \cdot e^{-j\omega t} dt \\
 &= \lim_{T_0 \rightarrow 0} \left(\int_{-T_0}^{T_0} \frac{1}{2T_0} \text{rect}\left(\frac{t}{2T_0}\right) \cdot e^{-j\omega t} dt \right) \\
 &= \lim_{T_0 \rightarrow 0} \left(\frac{1}{2T_0} \int_{-T_0}^{T_0} 1 \cdot e^{-j\omega t} dt \right) \\
 &= \lim_{T_0 \rightarrow 0} \frac{1}{2T_0} \cdot \frac{2 \cdot \sin(\omega T_0)}{\omega} \\
 &= \lim_{T_0 \rightarrow 0} \frac{\sin(\omega T_0)}{T_0 \omega} \\
 &= \lim_{T_0 \rightarrow 0} \frac{\omega \cdot \cos(\omega T_0)}{\omega} \\
 &= 1
 \end{aligned}$$

Task 3

$$\begin{aligned}
 \circ x(t) &= \sigma(t) \cdot U_0 \cdot e^{-\frac{1}{RC}t} \\
 \bullet X(f) &= \int_{-\infty}^{\infty} x(t) \cdot e^{-j2\pi f t} dt \\
 &= \int_{-\infty}^{\infty} \sigma(t) \cdot U_0 \cdot e^{-\frac{1}{RC}t} \cdot e^{-j2\pi f t} dt \\
 &= \int_0^{\infty} U_0 \cdot e^{-\frac{1}{RC}t} \cdot e^{-j2\pi f t} dt \\
 &= U_0 \int_0^{\infty} e^{-t \cdot \left(\frac{1}{RC} + j2\pi f\right)} dt \\
 &= \frac{U_0}{-\left(\frac{1}{RC} + j2\pi f\right)} \left[e^{-t \cdot \left(\frac{1}{RC} + j2\pi f\right)} \right]_0^{\infty} \\
 &= \frac{U_0}{\frac{1}{RC} + j2\pi f}
 \end{aligned}$$

Task 4

a)

$$x(t) = t \cdot \text{rect}\left(\frac{t}{2}\right) + \text{rect}\left(\frac{t}{2}\right)$$

b)

$$\circ x(t) = t \cdot \text{rect}\left(\frac{t}{2}\right) + \text{rect}\left(\frac{t}{2}\right)$$

$$\begin{aligned} \bullet X(j\omega) &= \int_{-\infty}^{\infty} x(t) \cdot e^{-j\omega t} dt \\ &= \int_{-\infty}^{\infty} \left(t \cdot \text{rect}\left(\frac{t}{2}\right) + \text{rect}\left(\frac{t}{2}\right) \right) \cdot e^{-j\omega t} dt \\ &= \int_{-1}^1 (t + 1) \cdot e^{-j\omega t} dt \\ &= \left[-\frac{t}{j\omega} e^{-j\omega t} - \frac{1}{j\omega} e^{-j\omega t} + \frac{1}{\omega^2} e^{-j\omega t} \right]_{-1}^1 \\ &= -\frac{2}{j\omega} e^{-j\omega} - \frac{1}{\omega^2} (e^{j\omega} - e^{-j\omega}) \\ &= 2j \left(-\frac{\sin(\omega)}{\omega^2} + \frac{\cos(\omega)}{\omega} \right) + 2\text{si}(\omega) \end{aligned}$$

Task 5

$$\bullet X(f) = \text{rect}(f)$$

$$\begin{aligned} \circ x(t) &= \int_{-\infty}^{+\infty} X(f) \cdot e^{j2\pi f t} df \\ &= \int_{-\infty}^{+\infty} \text{rect}(f) \cdot e^{j2\pi f t} df \\ &= \int_{-\frac{1}{2}}^{\frac{1}{2}} 1 \cdot e^{j2\pi f t} df \\ &= \frac{1}{j2\pi t} \left[e^{j2\pi f t} \right]_{-\frac{1}{2}}^{\frac{1}{2}} \\ &= \frac{1}{j2\pi t} (e^{j\pi t} - e^{-j\pi t}) \\ &= \frac{1}{\pi t} \cdot \sin(\pi t) \\ &= \text{si}(\pi t) \end{aligned}$$

Task 6

a) ---
b)

I) $x(t) = \text{rect}(t) * \text{rect}(t)$
 $X(f) = \text{si}^2\left(\frac{\omega}{2}\right)$
 $x(t) = \text{tri}(t)$

II) $x(t) = \sin(\pi t) * \delta\left(t - \frac{1}{2}\right)$
 $X(f) = j\pi(\delta(\omega + \pi) - \delta(\omega - \pi)) \cdot 1 \cdot e^{-j\omega \frac{1}{2}}$
 $x(t) = \sin\left(\pi\left(t - \frac{1}{2}\right)\right)$

c)

I) $x(t) = \frac{1}{2\pi} \int_{-\infty}^{+\infty} X(j\omega) \cdot e^{j\omega t} d\omega$
 $= \frac{1}{2\pi} \left(\int_{-11}^{-9} -2\pi j \cdot e^{j\omega t} d\omega + \int_{+9}^{+11} +2\pi j \cdot e^{j\omega t} d\omega \right)$
 $= \frac{1}{2\pi} \left(\left[\frac{-2\pi}{t} \cdot e^{j\omega t} \right]_{-11}^{-9} + \left[\frac{+2\pi}{t} \cdot e^{j\omega t} \right]_{+9}^{+11} \right)$
 $= \frac{1}{t} \left((e^{jt} - e^{-jt})(e^{10jt} - e^{-10jt}) \right)$
 $= -4 \cdot \sin(10t) \cdot \text{si}(t)$

II) $X_1(j\omega) = 2\pi j \cdot \text{rect}\left(\frac{\omega}{2}\right)$
 $X_2(j\omega) = \delta(\omega - 10) - \delta(\omega + 10)$

$X(j\omega) = X_1(j\omega) * X_2(j\omega)$
 $= 2\pi j \cdot \text{rect}\left(\frac{\omega}{2}\right) * (\delta(\omega - 10) - \delta(\omega + 10))$
 $= -2 \cdot \pi j (\delta(\omega + 10) - \delta(\omega - 10)) * \text{rect}\left(\frac{\omega}{2}\right)$
 $x(t) = -2 \cdot \sin(10t) \cdot \frac{1}{\pi} \text{si}(t) \cdot 2\pi$
 $= -4 \cdot \sin(10t) \cdot \text{si}(t)$

Task 7

1 – D	Triangle in the frequency domain corresponds to si^2 in the time domain Increasing phase of $4\pi/\omega_g$ causes shift to the left by $4t_1$ $\left(\frac{4\pi}{\omega_g} = 4t_1 \text{ with } t_1 = \frac{\pi}{\omega_g}\right)$
2 – C	Spectrum without symmetry causes complex-valued signal in the time domain
3 – F	Rectangle in the frequency domain corresponds to si in the time domain Phase $e^{-j\pi/2}$ corresponds to prefactor $-j$
4 – A	si^2 in the frequency domain corresponds to triangle in the time domain Phase equal to 0 means no shift in time
5 – B	Triangle in the frequency domain corresponds to si^2 in the time domain Phase equal to 0 means no shift in time
6 – E	Triangle in the frequency domain corresponds to si^2 in the time domain Decreasing phase of $4\pi/\omega_g$ causes shift to the right by $4t_1$ $\left(\frac{-4\pi}{\omega_g} = -4t_1 \text{ with } t_1 = \frac{\pi}{\omega_g}\right)$